KPPSC PMS Geography Paper-I — Q.8: Aeolian & Coastal Geomorphology

KPPSC PMS Master Examination Notes — Paper-I (Physical & General Geography)

Target Examination	KPPSC PMS (Provincial Management Service) — Khyber Pakhtunkhwa
Paper & Subject	Geography Paper-I (Physical & General Geography — 100 Marks)
EXAM QUESTIONS LINKED:	• **2016 Q.6:** "\"The coastal system is the scene of complex tidal fluctuation, winds, waves, and ocean currents.\" Discuss coastal processes and coastal landforms. (20 Marks)" • **2016 Q.8(c):** "Write short note on: Wind deposition / Aeolian landforms. (10 Marks)"
Target Syllabus	Section 2: Landforms — Wind (Aeolian) & Coastal Geomorphology
Marks & Allocation	20 Marks | Strict 35-Minute Descriptive Writing Limit
Core Theoretical Thesis	Aeolian and coastal geomorphic systems represent dynamic fluid-boundary environments governed by kinetic boundary-layer transport. In arid lands, wind operates via deflation and sandblasting abrasion, constructing distinctive erosional landforms (yardangs, zeugen, ventifacts) and organized depositional architectures (barchans, seifs, loess). In coastal zones, oscillating ocean waves, longshore drift, and tidal currents drive mechanical quarrying and hydraulic shock, carving erosional cliff-and-platform systems and depositing depositional tombolos, spits, and barrier bars, typified regionally by Pakistan's Makran Coast and Cholistan/Thar deserts.

🎯 Exam Questions Linked

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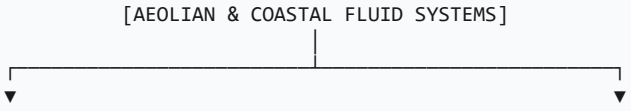
AEOLIAN DYNAMICS (BAGNOLD): Wind is a fluid of low density (1/800th of water); hence, aeolian erosion is restricted to arid/semi-arid regions lacking protective vegetative cover. Transport modes: Surface Creep (rolling of coarse grains 0.5–2 mm), Saltation (bouncing of sand 0.05–0.5 mm, accounting for 75–80% of total transport, restricted to <1–2 meters above ground), and Suspension (dust and silt <0.05 mm carried thousands of kilometers).

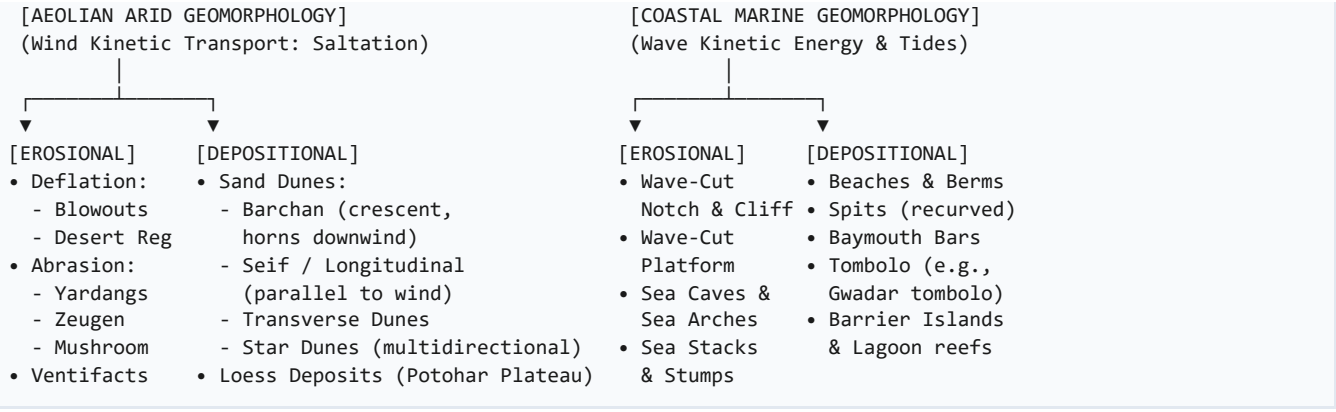
AEOLIAN LANDFORM TAXONOMY: Erosional: Deflation hollows/blowouts, Desert Pavement/Reg/Serir (armored surface left after fine dust is stripped), Ventifacts (faceted pebbles), Zeugen (tabular ridge-and-furrow ridges in horizontally bedded strata), Yardangs (elongated streamlined ridges in vertically tilted beds), Pedestal/Mushroom rocks. Depositional: Dunes (Barchans = crescentic, horns point downwind; Seifs = longitudinal linear ridges parallel to wind; Transverse = perpendicular to wind; Star dunes); Loess = unstratified, fertile, porous wind-blown silt blankets (e.g., Potohar Loess plateau in Pakistan).

COASTAL WAVE MECHANICS & LANDFORMS: Wave energy derives from offshore wind shear stress. Waves break as solitary waves in the surf zone, generating swash (upslope surge) and backwash (gravity return). Longshore drift (oblique wave approach) transports sediment parallel to shore.

COASTAL EROSIONAL VS. DEPOSITIONAL: Erosional: Wave-cut notch -> Sea cliff retreat -> Wave-cut platform -> Sea cave -> Sea arch -> Sea stack -> Sea stump. Depositional: Beaches, Spits (hook-shaped sand ridges extending into bays), Baymouth bars, Tombolos (sand spit connecting mainland to offshore island, e.g., Gwadar Peninsula hammerhead connected by a sandy isthmus tombolo).

1. Conceptual Mind Map & Thematic Roadmap





2. Analytical Dimensions & Geomorphic/Systemic Matrix

Geomorphic Realm / Process	Physical Transport & Fluid Mechanism	Diagnostic Diagnostic Landforms & Pakistan Field Manifestations
Aeolian Erosion: Deflation	Wind lifts and sweeps away loose, dry, silt-sized particles; leaves behind coarse, interlocking gravel armor.	Deflation hollows (e.g., Qattara Depression); Desert Pavements (Reg in Algeria, Serir in Libya, Dasht in Balochistan).
Aeolian Erosion: Abrasion	Sand grains saltating within 1 meter of ground act as sandblasters, cutting softer rock strata preferentially.	Yardangs (streamlined inverted ship-hull ridges), Zeugen (table-like ridges), Mushroom/Pedestal rocks; common in Chagai desert (Balochistan).
Aeolian Deposition: Barchans	Unidirectional wind, limited sand supply; sand cascades down steep slip-face (34° angle of repose).	Crescent-shaped sand dunes; convex windward slope, crescent horns pointing downwind; highly mobile across Cholistan and Thar deserts.
Aeolian Deposition: Loess	Suspended periglacial or desert dust carried vast distances, settling as unstratified, porous, calcareous silt blankets.	Vertical cliff-cleaving loess deposits; exceptional agricultural fertility, prone to badland gully erosion (e.g., Potohar Loess plateau in Rawalpindi/Jhelum).
Coastal Erosion: Cliff-Stack Sequence	Hydraulic shock pressure (up to 30,000 kg/m^2) and abrasion compress air into joints, excavating rock.	Wave-cut notch -> Cliff retreat -> Wave-cut platform -> Caves meeting to form Sea Arches -> Arch roof collapses leaving isolated Sea Stacks (e.g., Makran coast).
Coastal Deposition: Spits & Tombolos	Longshore drift carries sand along coast; sediment extends across bay mouth or ties island to mainland.	Sand spits, baymouth bars enclosing lagoons, and Tombolos (e.g., Gwadar Hammerhead Peninsula, where a tombolo connects an offshore bedrock block to Balochistan mainland).

3. 20-Mark Master Model Essay

Introduction: Boundary-Layer Geomorphic Systems in Arid and Marine Realms

Aeolian (wind) and coastal (marine wave) geomorphic systems operate as energetic fluid-boundary environments where the interface between Earth's lithosphere, atmosphere, and hydrosphere produces distinct landforms. In hyper-arid and semi-arid desert basins—which blanket nearly one-third of the global continental surface—liquid water is scarce, and vegetative cover is absent. Here, the turbulent boundary layer of atmospheric wind acts as a primary agent of sediment entrainment and landform evolution. Conversely, along the planet's 600,000 kilometers of coastline, oceanic kinetic energy—concentrated into breaking surf, nearshore waves, longshore drift, and tidal oscillations—sculpts the land-sea margin through relentless mechanical attack. Both geomorphic realms illustrate the principles of sediment fluid dynamics, where shear stress governs the threshold between erosion, transportation, and organized deposition, vividly showcased in Pakistan across the windswept sands of Cholistan and Thar and the wave-battered cliffs of the Makran Coast.

Aeolian Dynamics: Entrainment, Transportation Mechanics, and Desert Pavements

Because air possesses a density approximately 1/800th that of liquid water, wind is a low-density fluid incapable of mobilizing coarse gravels under normal atmospheric pressures. In his classical 1941 treatise 'The Physics of Blown Sand and Desert Dunes', Brigadier Ralph A. Bagnold established that aeolian transport occurs via three distinct mechanical

modes: Surface Creep, Saltation, and Suspension. Coarse sand grains (0.5 to 2.0 mm) roll along the desert floor under the impact of falling particles (surface creep, 20–25% of load). Fine dust, silt, and clays (<0.05 mm) are caught in turbulent thermal updrafts and carried thousands of kilometers aloft in suspension. However, the workhorse of aeolian transport is 'Saltation' (accounting for 75–80% of total sediment movement): medium sand grains (0.05 to 0.5 mm) are lifted by aerodynamic drag, bounce downwind in parabolic arcs, and strike the surface, ejecting further grains upon impact. Because gravitational settling confines saltating sand grains within 1 to 2 meters of the surface, wind abrasion is strictly a low-altitude phenomenon.

Aeolian Landforms: Erosional Yardangs and Depositional Sand Seas (Ergs)

Aeolian erosion operates through two physical processes: Deflation and Abrasion. 'Deflation' is the direct lifting and removal of unconsolidated dry sand and silt by wind turbulence. Over time, deflation hollows out vast desert depressions (such as the Qattara Depression in Egypt) and strips away all fine sediments, leaving behind an armored, interlocking mosaic of coarse pebbles and gravel known as a 'Desert Pavement' (termed 'Reg' in the Sahara and 'Dasht' in Balochistan). 'Abrasion' is the mechanical sandblasting of rock by saltating quartz grains. Differential abrasion carves 'Zeugen' (ridge-and-furrow terrain developed in horizontally bedded alternating hard and soft rocks), 'Yardangs' (elongated, aerodynamic, streamlined ridges sculpted in vertically tilted strata, aligned parallel to prevailing winds), and 'Mushroom Rocks' (pedestals undercut near their base). Depositional aeolian landforms divide into Sand Dunes and Loess. Dunes are organized sand accumulations classified by wind consistency and sand supply: 'Barchans' are crescent-shaped dunes formed under unidirectional winds and limited sand, whose convex face faces the wind while two horns point downwind; 'Seif' (or Longitudinal) dunes are knife-edge linear ridges running parallel to the resultant wind direction; 'Transverse' dunes form wavy sand ridges perpendicular to strong winds; and 'Star Dunes' develop under multidirectional wind regimes. Beyond deserts, wind deposits 'Loess'—extensive, unstratified blankets of calcareous silt blown from glacial outwash or deserts, epitomized in Pakistan by the Potohar Loess Plateau.

Coastal Wave Mechanics: Nearshore Energy Dissipation and Longshore Transport

In the marine environment, the primary geomorphic engine is the wind-generated oceanic wave. As deep-water oscillatory waves (where water molecules move in closed circular orbits) approach a sloping shoreline, water depth decreases to less than half the wavelength ($d < L/2$). Friction with the seabed slows wave velocity, compresses wavelength, and steepens wave height until the wave becomes unstable and breaks violently in the surf zone. This transformation converts potential energy into intense kinetic energy: the breaking wave surges up the beach as 'swash' and retreats down-slope under gravity as 'backwash'. Waves hitting the coast at an oblique angle generate a net current parallel to the shoreline known as 'Longshore Drift', which acts as an immense aquatic conveyor belt transporting millions of cubic meters of sand along continental margins.

The Coastal Erosional Sequence: From Wave-Cut Notches to Isolated Stacks

Coastal erosion proceeds through four aggressive mechanical processes: Hydraulic Action (where waves crashing against cliffs generate hydraulic shock pressures exceeding 30,000 kg/m^2), Cavitation (the explosive compression and decompression of air trapped in rock fissures), Abrasion/Corrasion (the battering of cliffs using sand and gravel as artillery), and Attrition. The coastal erosional sequence follows a predictable evolutionary path: waves relentlessly attack the base of a high sea cliff, excavating a horizontal 'Wave-Cut Notch'. As the notch deepens, the unsupported cliff overhang collapses under gravity, causing the sea cliff to retreat landward. This cliff recession leaves behind a gently sloping, bare bedrock platform at low-tide level known as a 'Wave-Cut Platform'. Along structurally heterogeneous headlands, wave refraction focuses hydraulic energy on rocky promontories. Waves quarry intersecting joints to excavate 'Sea Caves'. When waves breach through the headland from both sides, they unite to form a 'Sea Arch'. Eventually, gravitational stress and wave pounding collapse the arch roof, leaving an isolated, vertical bedrock pillar standing offshore known as a 'Sea Stack'. Continued abrasion reduces the stack to a submerged 'Sea Stump'.

Coastal Depositional Architecture and the Gwadar Tombolo Case Study

Where wave energy attenuates in sheltered bays or shallow waters, coastal deposition constructs constructive landforms. Foremost is the 'Beach'—an accumulation of wave-sorted sand, shingle, or shell fragments organized into berms and beach cusps. When longshore drift transports sand past a sharp change in coastline orientation (such as an inlet or bay mouth), wave energy drops, causing sand to deposit as a linear finger projecting into open water, known as a 'Spit' (often recurved at its tip by wave refraction). If a spit extends entirely across the mouth of a bay, it seals it off as a 'Baymouth Bar', enclosing a brackish coastal lagoon. A globally iconic coastal depositional landform is the 'Tombolo'—a

depositional spit or sandbar that links an offshore bedrock island to the mainland. The definitive textbook example in Pakistan is the 'Gwadar Peninsula': Gwadar was originally an isolated, hammerhead-shaped offshore rocky island in the Arabian Sea; over geological epochs, convergent wave refraction and longshore drift deposited a monumental, 12-kilometer-long sandy isthmus tombolo, permanently welding the Gwadar bedrock ridge to the mainland of Balochistan. In evaluative summary, aeolian and coastal geomorphology illustrate the supreme organizing capacity of fluid currents, structuring the arid basins and marine frontiers that sustain human transport, maritime trade, and coastal urbanization.

4. Scholarly Perspectives & Geomorphic Axioms

Scholar / Geomorphologist	Theoretical Contribution / Concept	Exact Quotable Academic Thesis / Axiom
R.A. Bagnold (1941)	The Physics of Blown Sand and Desert Dunes	"*Saltation is the primary engine of sand transport; sand grains moving in leaping trajectories govern both the erosion of bedforms and the geometric morphology of sand dunes.*"
Douglas W. Johnson (1919)	Shore Processes and Shoreline Development	"*Coastal profiles represent an evolutionary sequence governed by wave attack, progressing from initial notched headlands to mature, graded shore platforms and baymouth bars.*"
C.A.M. King (1972)	Beaches and Coasts	"*The morphology of any shoreline reflects a dynamic equilibrium between wave energy inputs, tidal ranges, and the availability of mobile littoral sediments.*"
Arthur N. Strahler (1969)	Physical Geography	"*In arid lands devoid of vegetal cohesion, wind and ephemeral flash floods combine to produce exceptional landforms characterized by desert pavements, sand seas, and pediments.*"
E.C.F. Bird (2008)	Coastal Geomorphology: An Introduction	"*Longshore drift and wave refraction operate as self-regulating hydrodynamic systems that sculpt sand spits, barrier chains, and connecting tombolos across continental shelves.*"